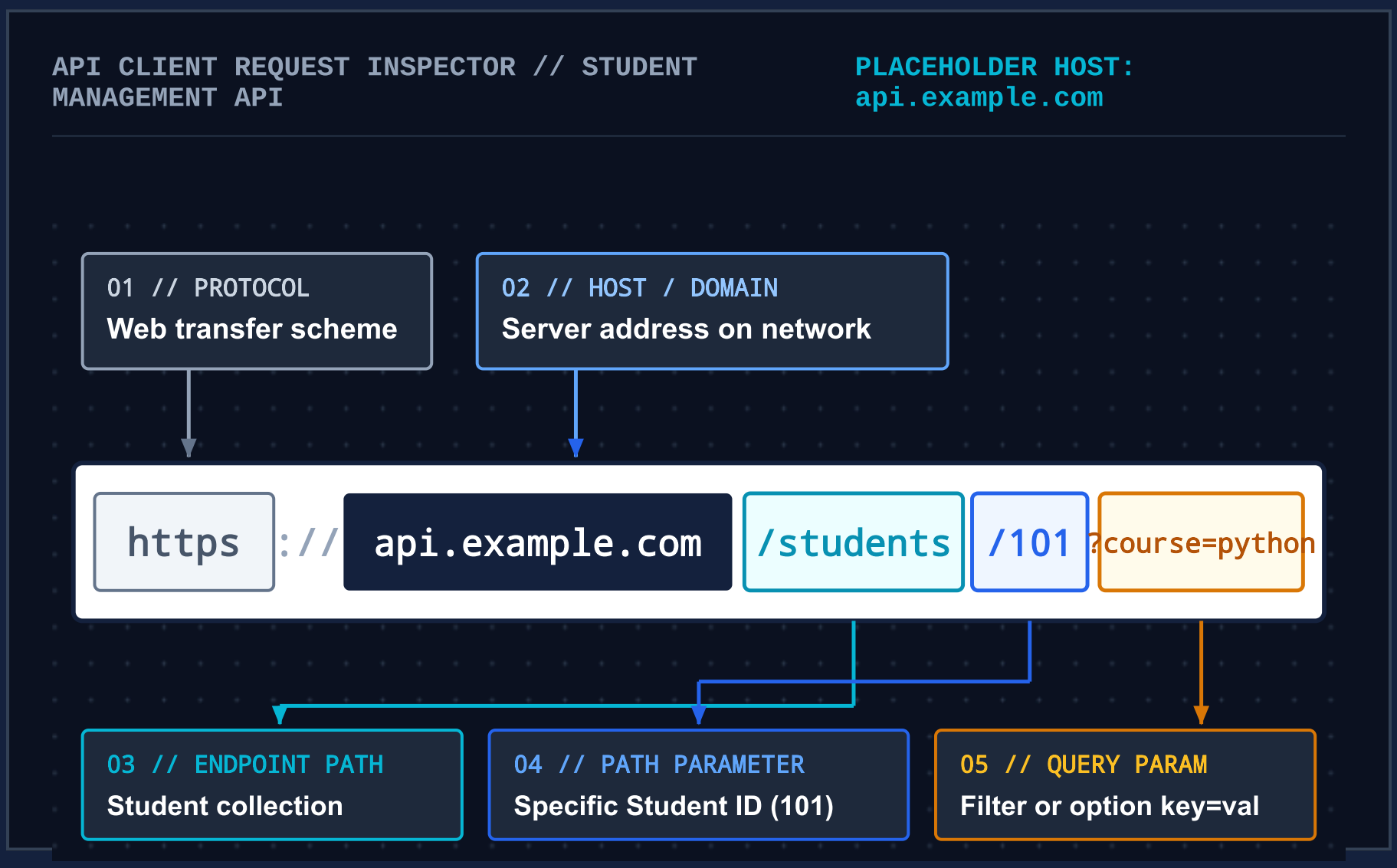
API ADDRESSING & ROUTING ARCHITECTURE

UNDERSTANDING API URLS

> “Endpoints, path parameters and query parameters explained.”



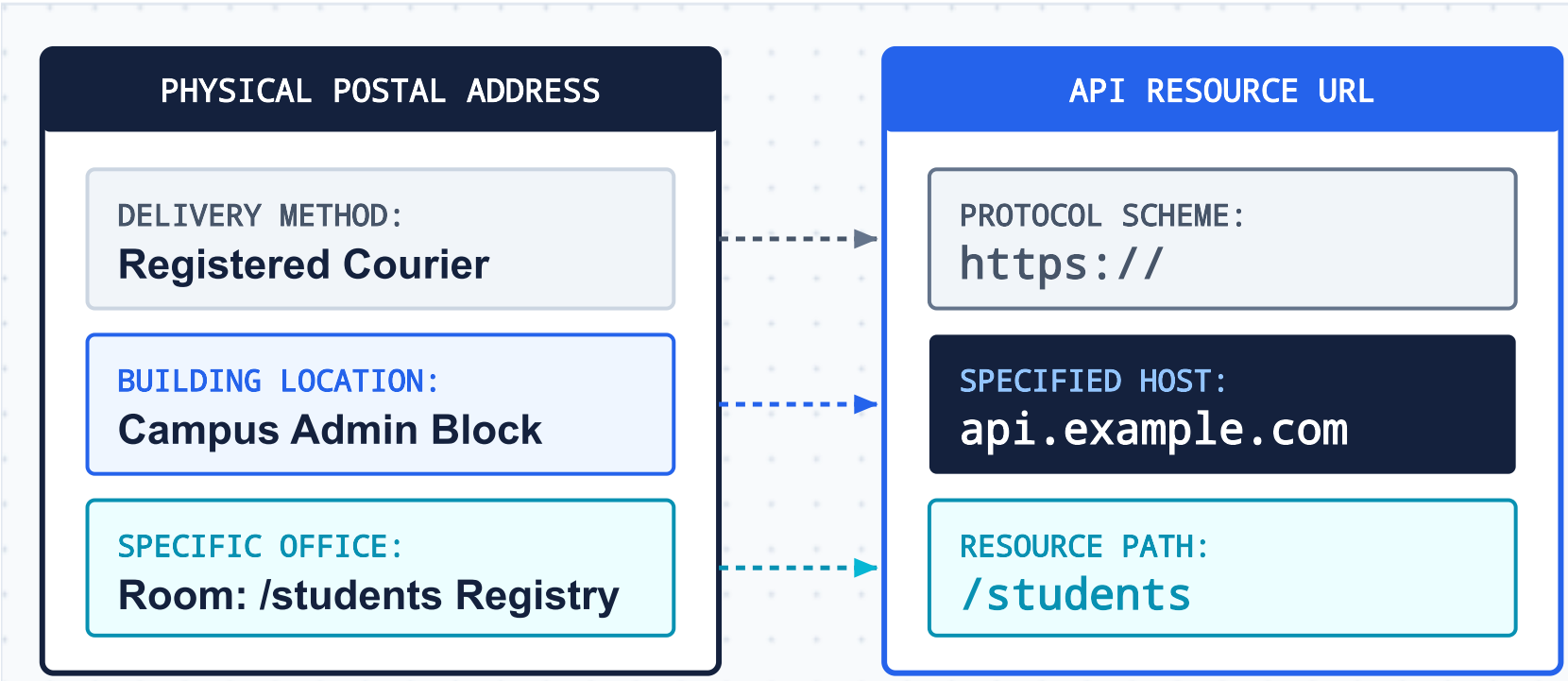
RUNNING CASE STUDY	CORE FOCUS	IMPLEMENTATION
Student Management API	Path vs. Query Params	Python 3.10+ & FastAPI

LOCATING RESOURCES ON A NETWORK

Every API request needs a destination.

A **URL (Uniform Resource Locator)** is an address that identifies where a resource can be accessed on a network.

FIG 05.1 // STRUCTURAL ANALOGY: POSTAL ADDRESS VS. API URL DESTINATION MAPPING



COMPLETE RESOURCE URL (PLACEHOLDER DOMAIN)

`https://api.example.com/students`

STUDENTS RESOURCE

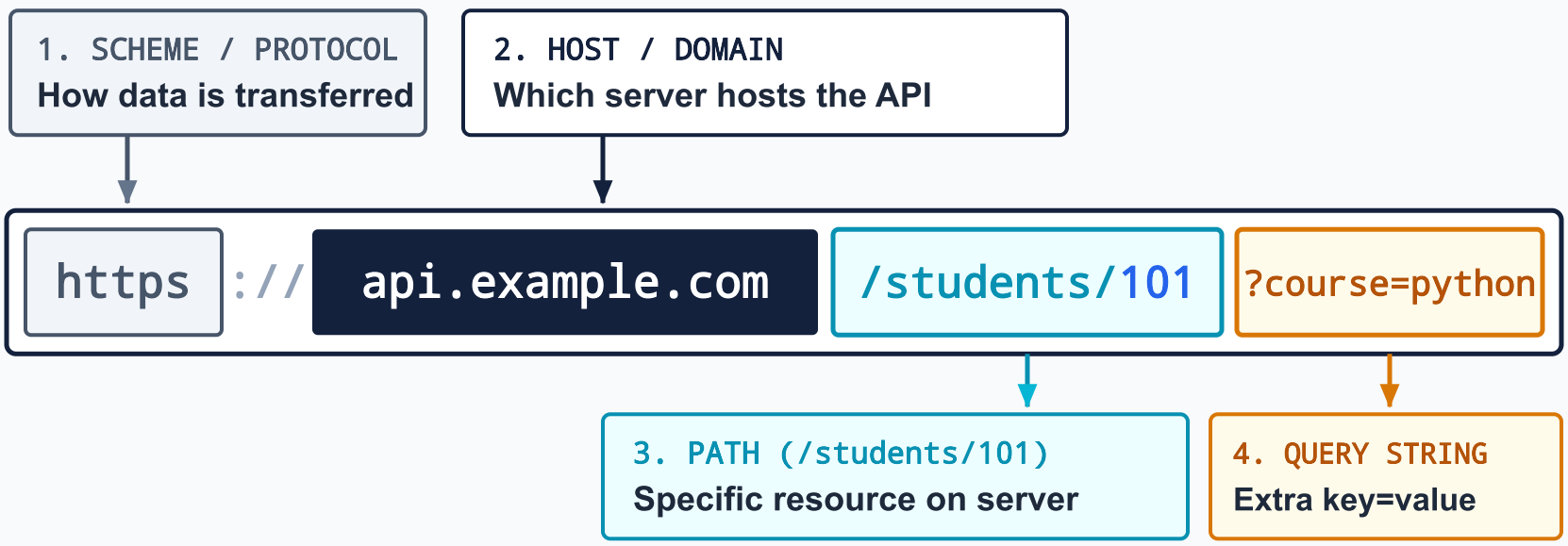
A URL tells the client where to send its request. However, the URL alone does not determine the action—the **HTTP method** (`GET`, `POST`, `DELETE`) specifies what to do there.

STRUCTURAL BREAKDOWN

A URL is made of different parts.

FIG 05.2 // COLOR-CODED ANATOMY OF AN API URL

4 STRUCTURAL ZONES



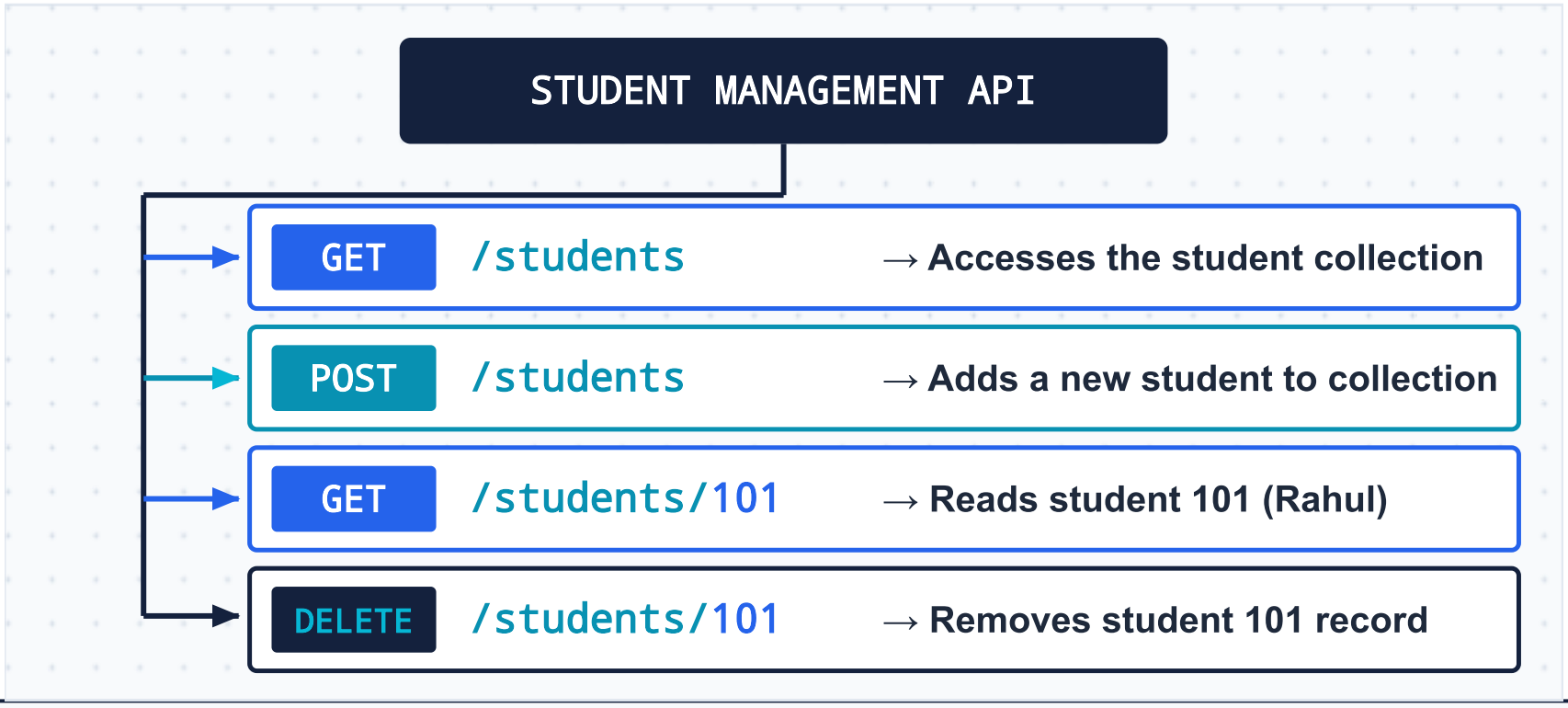
COMPONENT	EXAMPLE	WHAT IT REPRESENTS
Scheme / protocol	`https`	The communication protocol used to transfer data securely.
Host / domain	`api.example.com`	The domain name or address of the server hosting the API.
Path	`/students/101`	The hierarchical route pointing to the requested resource.
Query string	`?course=python`	Optional key-value parameters appended after `?`.

ROUTES & OPERATIONS

So, what exactly is an endpoint?

An **endpoint** is a specific API access point, commonly described by its **URL path** and the **HTTP method** used with it.

FIG 05.3 // STUDENT MANAGEMENT API ROUTE MAP — HTTP METHOD + PATH = OPERATION



COLLECTION VS. ITEM PATHS

`/students` targets the collection; `/students/101` identifies a particular student.

TECHNICAL DISTINCTION

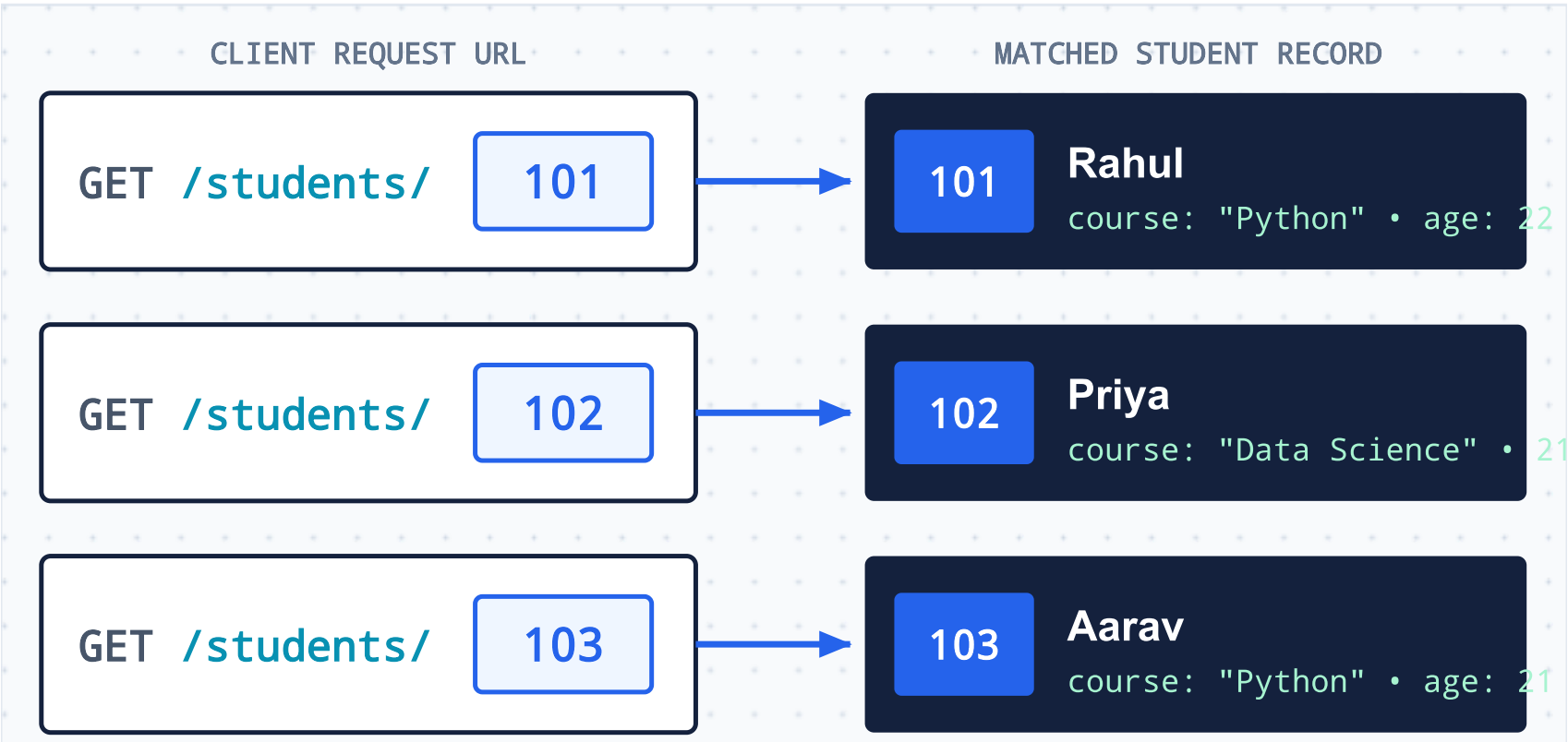
An endpoint is not always a full URL—developers often use it to mean a **route + HTTP method**.

DYNAMIC RESOURCE IDENTIFICATION

Path parameters identify a specific resource.

A **path parameter** is a variable value embedded directly inside the URL path. The path structure (`/students/`) stays constant while the ID changes:

FIG 05.4 // VARIABLE STUDENT ID ROUTING — STATIC PREFIX + DYNAMIC ID



Path parameter = Which specific resource? `/students/{id}`

PYTHON IMPLEMENTATION

How do we create a path parameter in FastAPI?

main.py — FastAPI Path Parameter Extraction

PYTHON 3.10+

REQUEST: GET /students/ 101

```python
from fastapi import FastAPI
app = FastAPI()

@app.get("/students/{student_id}")
def get_student(student_id: int):
    return {"student_id": student_id}
```

1. URL EXTRACTION
Captures "101" from path

2. TYPE CONVERSION
Converts to integer 101

RESPONSE (200 OK): {"student_id": 101}

01 // {student_id} IN ROUTE
Declares a dynamic path variable inside the route template.

02 // AUTOMATIC EXTRACTION
FastAPI reads the value from the URL when a request arrives.

03 // student_id: int
Validates and converts the string "101" into a Python int.

04 // FUNCTION ARGUMENT
Your Python function receives the ready-to-use integer 101.

FILTERING & CUSTOMIZING REQUESTS

Query parameters help filter or customize a request.

Query parameters are additional `key=value` pairs appended after the `?` in a URL.

FIG 05.5 // STUDENT DIRECTORY FILTER DASHBOARD — UI FILTERS → URL QUERY STRING



`GET /students?course=python`	Filter by course
`GET /students?age=21`	Filter by age
`GET /students?course=python&age=21`	Combined filters

Query parameter = What conditions or options? `?key=value`

AUTOMATIC QUERY INFERENCE

How do query parameters work in FastAPI?

Function parameters that are **not** part of the route path (`"/students"`) are automatically interpreted as **query parameters**:

main.py — Optional Query Parameters (Python 3.10+ Union Syntax) QUERY EXTRACTION

```
GET /students?course=python&age=21

@app.get("/students")
def get_students(
    course: str | None = None,
    age: int | None = None
):
    return {"course": course, "age": age}
```

course = "python"
String or None (Optional)

age = 21
Converted to int or None

RETURNED JSON: {"course": "python", "age": 21}

`str | None`
Python 3.10+ union type: value can be a `str` or `None`.

`= None`
Sets the default to `None`, making the parameter optional.

FLEXIBLE CALLS
Clients can pass one, both, or neither query parameter.

Note: This code demonstrates how FastAPI extracts and validates query parameters—actual database filtering logic is added inside the function body.

ARCHITECTURAL CONVENTION

They look similar. But they serve different purposes.

01 // PATH PARAMETER

Identify a Specific Resource

```
GET /students/101
```

COMMON EXAMPLES:

- Student ID (`/students/101`)
- Product ID (`/products/42`)
- Order ID (`/orders/9001`)

02 // QUERY PARAMETER

Filter or Customize Request

```
GET /students?course=python
```

COMMON EXAMPLES:

- Course or Age filter
- Sorting (`?sort=name`)
- Pagination (`?page=2&limit=20`)

PATH PARAMETER	QUERY PARAMETER
Embedded directly as part of the URL path	Appears at the end of the URL after `?`
Often identifies a single specific resource	Often filters, sorts, or modifies a collection request
`Example: /students/101`	`Example: /students?course=python`

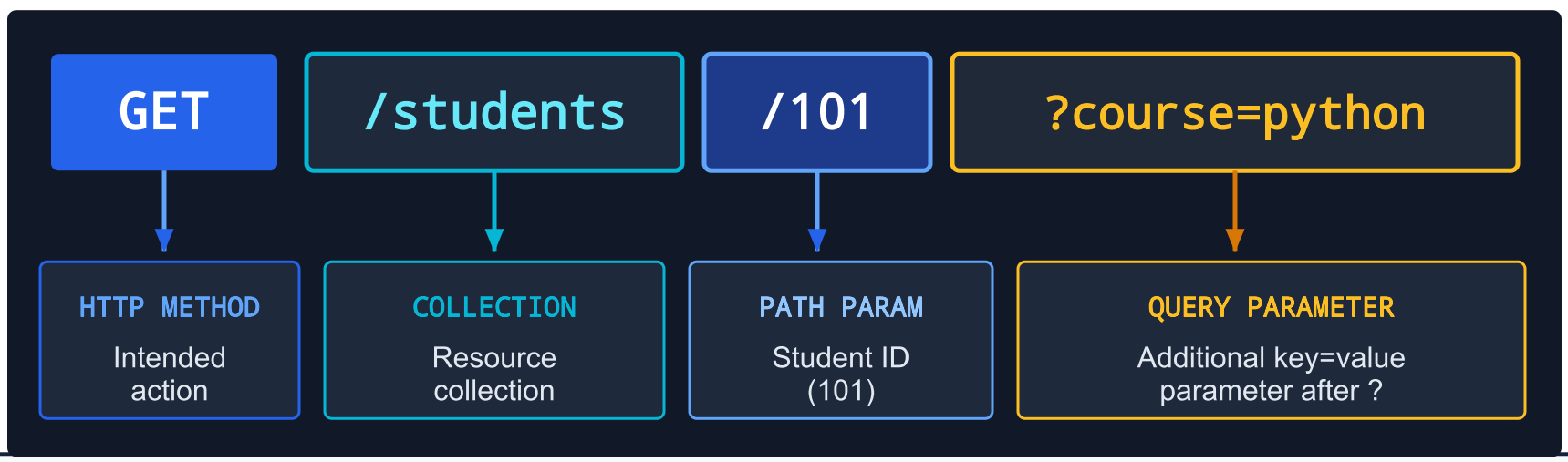
Design Perspective: Using path parameters for resource IDs and query parameters for filtering is the standard **REST API convention**, though exact usage always depends on an API's specific design.

READING AN API REQUEST AT A GLANCE

You can now read an API URL.

FIG 05.6 // COMPLETE ANNOTATED API REQUEST

METHOD + PATH + PARAMETERS



URL
Address of a resource.

ENDPOINT
API operation or access point.

PATH PARAMETER
Value inside the path (`/101`).

QUERY PARAMETER
Value after `?` (`course=python`).

COMING NEXT // PART 06 — BUILDING YOUR FIRST FASTAPI ENDPOINTS

“We'll turn these concepts into actual Python code and create working API routes.”

Save this carousel if you're learning backend development.